SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 0603 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

SCENARIO – BASED QUESTIONS

Code of Conduct:

I Mohammed Mudhassir A/192424367 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Mudhassir

1. Functional Blocks of a Machine Learning System (5 Marks)

A Machine Learning (ML) system consists of several functional blocks that work together to build a predictive model.

Data Collection:
The first step is collecting data from sources such as databases, sensors, websites, or IoT devices. The quality of the collected data directly affects the model's performance.

Data Preprocessing:
The collected data is cleaned by removing duplicate records, handling missing values, correcting inconsistencies, and converting categorical data into numerical form. The data is also normalized or standardized to improve learning efficiency.

Feature Extraction:
Important features that influence the prediction are selected from the dataset. Feature extraction reduces unnecessary information and improves model accuracy.

Model Training:
The processed data is used to train a machine learning algorithm such as Decision Tree, Neural Network, or Support Vector Machine. During training, the model learns patterns and relationships within the data.

Model Evaluation:
The trained model is tested using unseen data. Performance is measured using evaluation metrics such as Accuracy, Precision, Recall, and F1-Score.

Supervised Learning Example

In supervised learning, the dataset contains **labelled data**. For example, a spam email detection system is trained using emails labelled as "Spam" or "Not Spam". The model learns from these labels and predicts the class of new emails.

Unsupervised Learning Example

In unsupervised learning, the dataset contains **unlabelled data**. For example, customer segmentation groups customers with similar purchasing behavior without predefined labels.

Conclusion:
A machine learning system follows a sequence of data collection, preprocessing, feature extraction, model training, and evaluation to produce accurate predictions using labelled or unlabelled data.

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2. Machine Learning System for Disaster Management (5 Marks)

A machine learning system can predict natural disasters such as floods and earthquakes by analyzing historical and real-time environmental data.

Data Sources

- Historical rainfall records

- River water levels
- Seismic sensor readings
- Weather forecasts
- Satellite images
- Temperature and humidity data

Data Preprocessing

- Remove missing and duplicate values.
- Normalize numerical features.
- Encode categorical variables.
- Detect and remove outliers.
- Split the dataset into training and testing sets.

Feature Selection

Important features include:

- Rainfall intensity
- Water level
- Soil moisture
- Temperature
- Seismic activity
- Atmospheric pressure

Selecting relevant features reduces computation and improves prediction accuracy.

Model Selection

A **Neural Network (ANN)** is suitable because it can learn complex nonlinear relationships from large datasets. For flood prediction, LSTM (Long Short-Term Memory) networks are also effective because they handle time-series data efficiently.

Training and Validation

The model is trained using historical disaster data. Validation data is used to tune hyperparameters and avoid overfitting. Testing data evaluates the final model performance.

Evaluation

Performance is measured using:

- Accuracy
- Precision

- Recall
- F1-Score

Applications

The system provides:

- Early flood and earthquake warnings.
- Risk assessment.
- Evacuation planning.
- Disaster response support.
- Resource allocation for emergency services.

Conclusion:
Machine learning enables accurate disaster prediction, reducing loss of life and property through timely warnings and effective emergency planning.

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3. Backpropagation in a Multilayer Perceptron (5 Marks)

Backpropagation is a supervised learning algorithm used to train neural networks by updating weights to minimize prediction errors.

Forward Pass

Input values are passed through the input layer, hidden layer, and output layer. Each neuron computes a weighted sum followed by an activation function such as Sigmoid or ReLU to produce the output.

Error Calculation

The predicted output is compared with the actual output using a loss function.

Error = Actual Output − Predicted Output

Backward Pass

The error is propagated backward through the network. Gradients are calculated, and weights are updated using gradient descent.

Weight Update Formula

New Weight = Old Weight − Learning Rate × Gradient

Example

Suppose:

- Actual Output = 1
- Predicted Output = 0.8

Error = 1 − 0.8 = 0.2

The weights are adjusted to reduce this error during the next iteration.

Effect of Learning Rate

- A very high learning rate causes unstable learning.
- A very low learning rate makes training slow.
- An appropriate learning rate ensures faster convergence.

Effect of Activation Functions

- **Sigmoid** is suitable for binary classification.
- **ReLU** trains faster and avoids the vanishing gradient problem.
- **Softmax** is used for multi-class classification.

Conclusion:
Backpropagation continuously updates network weights to minimize prediction error and improve classification accuracy.

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4. Genetic Algorithm Population Evolution (5 Marks)

Dataset

Example	Speed	Accuracy	Fitness	Suitable
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

Selection

Selection chooses chromosomes with higher fitness values.

Selected parents:

- Example 5 (Fitness = 80)
- Example 3 (Fitness = 75)

These solutions have better performance and are more likely to produce good offspring.

Crossover

The selected parents exchange characteristics.

Parent 1:

- Medium Speed
- High Accuracy

Parent 2:

- High Speed
- Medium Accuracy

Offspring:

- High Speed
- High Accuracy

This new solution combines the strengths of both parents.

Mutation

Mutation introduces small random changes.

Example:

- Speed changes from Medium to High.
- Accuracy changes from Medium to High.

Mutation increases diversity and helps avoid local optimum solutions.

Evolution

Generation 1

Fitness:
40, 60, 75, 65, 80

↓

Selection

↓

Crossover

↓

Mutation

↓

Generation 2

Fitness:
65, 78, 82, 85

The average fitness improves after evolution.

Best Optimized Solution

Speed	Accuracy	Fitness
High	High	85

Conclusion:
Selection chooses the best individuals, crossover combines good characteristics, and mutation introduces diversity, resulting in an optimized solution with the highest fitness value.

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5. Neural Network Classification and Performance Evaluation (5 Marks)

Dataset

Example	Actual	Predicted
1	Yes	Yes
2	Yes	No
3	No	No
4	Yes	Yes
5	No	Yes

Backpropagation Training

The neural network receives input features such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. During training, the predicted output is compared with the actual output. The error is propagated backward, and the weights are updated repeatedly until the prediction error is minimized.

Confusion Matrix Values

- **True Positive (TP)** = 2 (Examples 1 and 4)
- **True Negative (TN)** = 1 (Example 3)
- **False Positive (FP)** = 1 (Example 5)
- **False Negative (FN)** = 1 (Example 2)

Performance Metrics

Accuracy

= (TP + TN) / Total

= (2 + 1) / 5

= **60%**

Precision

= TP / (TP + FP)

= 2 / (2 + 1)

= **66.67%**

Recall

= TP / (TP + FN)

= 2 / (2 + 1)

= **66.67%**

F1-Score

= 2 × (Precision × Recall) / (Precision + Recall)

= **66.67%**

Importance of Preprocessing

- Removes missing values.
- Normalizes numerical features.
- Removes noisy data.
- Improves model accuracy.

Importance of Feature Selection

Selecting important features such as Glucose Level and Blood Pressure reduces computation, avoids overfitting, and improves prediction reliability.

Conclusion:
The neural network correctly classified most patient records, achieving **60% accuracy**. Proper preprocessing and feature selection significantly improve the model's performance and reliability.

6. Architecture of a Decision Tree Learning System (5 Marks)

A **Decision Tree** is a supervised machine learning algorithm used for classification and regression. It classifies data by splitting it into smaller subsets based on the most informative features.

Architecture of a Decision Tree

A decision tree consists of:

- **Root Node:** Represents the entire dataset and the first decision.
- **Decision Nodes:** Represent tests on input features.
- **Branches:** Represent possible outcomes of each test.
- **Leaf Nodes:** Represent the final class or prediction.

Feature Selection

The best feature is selected using **Entropy** and **Information Gain**.

Entropy: Measures the impurity or uncertainty in the dataset. Lower entropy indicates purer data.

Information Gain: Measures the reduction in entropy after splitting the dataset. The feature with the highest information gain is selected as the root node.

Example

Consider a dataset with features **Weather** and **Play Cricket**.

Weather	Play Cricket
Sunny	No
Rainy	Yes
Cloudy	Yes
Sunny	No
Rainy	Yes

The feature **Weather** has the highest information gain, so it becomes the root node.

Tree Construction

1. Calculate entropy.
2. Compute information gain for each feature.
3. Select the feature with the highest information gain.
4. Split the dataset.
5. Repeat until all records belong to the same class or stopping criteria are met.

Pruning

Pruning removes unnecessary branches from the decision tree to reduce complexity and improve generalization.

Controlling Overfitting

Overfitting can be controlled by:

- Limiting tree depth.
- Setting a minimum number of samples per node.
- Applying pre-pruning or post-pruning.
- Using cross-validation.

Conclusion:
Decision Trees classify data by selecting features with the highest information gain. Pruning helps reduce overfitting and improves prediction accuracy.

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7. Machine Learning System for Smart Traffic Management (5 Marks)

A smart traffic management system predicts traffic congestion using machine learning and real-time traffic data.

Data Collection

The system collects:

- Vehicle density
- Time of day
- Weather conditions
- Road type
- GPS location
- Traffic camera data

Data Preprocessing

- Remove missing values.
- Normalize numerical data.
- Encode categorical variables.
- Remove duplicate records.
- Split data into training and testing sets.

Feature Engineering

Important features include:

- Peak hour
- Average vehicle speed
- Traffic density
- Rainfall
- Road category

These features improve prediction performance.

Algorithm Selection

A **Random Forest** or **Artificial Neural Network (ANN)** is suitable because they can model complex traffic patterns and provide high prediction accuracy.

Training and Validation

The model is trained using historical traffic data. Validation data is used to tune parameters and prevent overfitting. The testing dataset evaluates final performance.

Deployment

The trained model is integrated into traffic control systems to:

- Predict congestion.
- Suggest alternative routes.
- Optimize traffic signals.
- Reduce travel time.

Conclusion:
Machine learning improves traffic management by predicting congestion accurately, reducing delays, and supporting smart city development.

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8. Convolutional Neural Networks (CNNs) (5 Marks)

A **Convolutional Neural Network (CNN)** is a deep learning model designed for image classification and computer vision tasks.

Layers in CNN

Convolution Layer

- Applies filters to detect features such as edges, textures, and shapes.
- Produces feature maps.

Pooling Layer

- Reduces feature map size.
- Decreases computation.
- Helps prevent overfitting.

Fully Connected Layer

- Connects extracted features to output neurons.
- Performs final classification.

Feature Map Generation

Each convolution filter scans the image and extracts important visual features. Multiple filters generate multiple feature maps representing different characteristics.

Backpropagation

The prediction error is calculated, and weights are updated using gradient descent to minimize the loss function.

Example

A CNN classifies images of cats and dogs.

- Convolution detects ears, eyes, and fur.
- Pooling reduces image size.
- Fully connected layer predicts "Cat" or "Dog."

Factors Affecting Performance

- Dataset quality.
- Number of training images.
- Learning rate.
- Number of convolution layers.
- Batch size.
- Activation functions.
- Epochs.

Conclusion:
CNNs automatically learn image features and provide high accuracy in image classification tasks.

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9. Genetic Algorithms for Feature Selection (5 Marks)

Feature selection identifies the most relevant attributes for building an efficient machine learning model.

Chromosome Representation

Each chromosome is represented as a binary string.

Example:

Features:
Age, BP, Glucose, BMI, Cholesterol

Chromosome:
1 0 1 1 0

Here,

- 1 = Feature Selected
- 0 = Feature Not Selected

Selected features:
Age, Glucose, BMI

Fitness Evaluation

Each chromosome is evaluated using model accuracy.

Higher accuracy means higher fitness.

Selection

The best chromosomes with higher fitness values are selected for reproduction.

Crossover

Selected chromosomes exchange genes to produce new feature subsets.

Mutation

Random bits are changed from 0 to 1 or 1 to 0 to introduce diversity.

Benefits

- Removes irrelevant features.
- Improves prediction accuracy.
- Reduces training time.
- Prevents overfitting.
- Produces simpler models.

Conclusion:
Genetic Algorithms efficiently identify optimal feature subsets, improving both accuracy and computational efficiency.

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10. Evaluation Metrics for Classification Models (5 Marks)

Evaluation metrics measure the performance of classification models.

Sample Confusion Matrix

	Predicted Positive	**Predicted Negative**
Actual Positive	TP = 50	FN = 10
Actual Negative	FP = 5	TN = 35

Accuracy

Accuracy = (TP + TN) / Total

= (50 + 35) / 100

= **85%**

Precision

Precision = TP / (TP + FP)

= 50 / (50 + 5)

= **90.91%**

Precision measures how many predicted positive cases are actually positive.

Recall

Recall = TP / (TP + FN)

= 50 / (50 + 10)

= **83.33%**

Recall measures how many actual positive cases are correctly identified.

F1-Score

F1 = 2 × (Precision × Recall) / (Precision + Recall)

= **86.96%**

F1-score balances Precision and Recall.

ROC Curve

The ROC (Receiver Operating Characteristic) curve plots the True Positive Rate against the False Positive Rate. A model with a larger Area Under the Curve (AUC) has better classification performance.

Importance

Medical Diagnosis

- High Recall is important to detect all disease cases.

Fraud Detection

- High Precision reduces false fraud alerts.

Conclusion:
Accuracy, Precision, Recall, F1-score, and ROC-AUC provide a comprehensive evaluation of classification models, helping select the most suitable model for real-world applications.

11. Architecture of an Artificial Neural Network (ANN) (5 Marks)

An **Artificial Neural Network (ANN)** is a machine learning model inspired by the structure of the human brain. It consists of interconnected neurons organized into layers that process information and make predictions.

Components of ANN

Input Layer

- Receives the input features from the dataset.
- Example: Age, Blood Pressure, Glucose Level, and Lifestyle Risk.

Hidden Layer

- Performs mathematical computations using weights, biases, and activation functions.
- Learns complex relationships between input and output.

Output Layer

- Produces the final prediction.
- Example: Disease = Yes or No.

Weights and Biases

- **Weights** determine the importance of each input feature.
- **Bias** shifts the activation function to improve prediction accuracy.

Forward Propagation

1. Input data is received by the input layer.
2. Inputs are multiplied by weights and added with bias.
3. The activation function (ReLU or Sigmoid) is applied.
4. The output is passed to the next layer.
5. The final output is generated.

Example

Inputs:

- Age = 45
- Blood Pressure = High
- Glucose = High

The ANN processes these inputs and predicts:

Disease = Yes

Controlling Overfitting

Overfitting occurs when the model memorizes training data instead of learning general patterns.

Techniques to prevent overfitting include:

- **Dropout:** Randomly disables some neurons during training.
- **Regularization (L1/L2):** Penalizes large weight values.
- **Early Stopping:** Stops training when validation error increases.

- **Cross Validation:** Uses multiple data splits for better generalization.

Conclusion

ANNs learn complex patterns from data using forward propagation and weight updates. Techniques like dropout and regularization improve model generalization and prediction accuracy.

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12. Machine Learning System for Weather Forecasting (5 Marks)

A machine learning system can predict rainfall using weather-related features such as temperature, humidity, wind speed, and atmospheric pressure.

Data Collection

The dataset includes:

- Temperature
- Humidity
- Wind Speed
- Atmospheric Pressure
- Rainfall History
- Cloud Cover

Data Preprocessing

- Remove missing values.
- Normalize numerical data.
- Remove duplicate records.
- Handle outliers.
- Split the dataset into training and testing sets.

Feature Engineering

Important features include:

- Temperature
- Humidity
- Wind Speed
- Air Pressure
- Rainfall in previous days

Selecting relevant features improves model performance.

Algorithm Selection

A **Random Forest** or **Artificial Neural Network (ANN)** is suitable because they can model complex weather relationships and provide accurate predictions.

Model Training

The model learns patterns from historical weather data.

Validation

Validation data is used to optimize model parameters and avoid overfitting.

Evaluation

Performance metrics include:

- Accuracy
- Precision
- Recall
- Mean Squared Error (for regression)

Deployment

The trained model is deployed to weather forecasting systems where it predicts rainfall in real time using live weather sensor data.

Applications

- Agriculture
- Flood prediction
- Disaster management
- Aviation
- Water resource planning

Conclusion

Machine learning improves rainfall prediction accuracy and helps governments and farmers make informed decisions.

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13. Recurrent Neural Networks (RNNs) (5 Marks)

A **Recurrent Neural Network (RNN)** is a deep learning model specially designed to process sequential or time-series data.

Unlike traditional neural networks, RNNs remember previous information using **hidden states**.

Architecture

The RNN processes one input at a time while maintaining information from previous time steps.

Hidden State

The hidden state stores information from previous inputs and passes it to the next time step.

This enables the network to learn dependencies in sequences.

Working Process

1. Input sequence is provided.
2. Hidden state stores previous information.
3. Current input and previous hidden state are combined.
4. Output is generated.
5. Hidden state is updated for the next input.

Example

Text Prediction

Input:
"I love machine"

Output:
"learning"

The RNN predicts the next word using previous words.

Time-Series Forecasting

An RNN predicts future stock prices or rainfall based on previous observations.

Challenges

Vanishing Gradient Problem

Gradients become very small during training, making it difficult to learn long-term dependencies.

Solutions

- Long Short-Term Memory (LSTM)
- Gated Recurrent Unit (GRU)
- Gradient Clipping

These techniques help preserve important information over long sequences.

Applications

- Speech Recognition
- Language Translation

- Weather Forecasting
- Stock Price Prediction
- Sentiment Analysis

Conclusion

RNNs effectively process sequential data by maintaining hidden states. Advanced variants such as LSTM and GRU overcome the limitations of standard RNNs.

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14. Genetic Algorithms for Task Scheduling (5 Marks)

A **Genetic Algorithm (GA)** is an optimization technique inspired by natural selection. It is widely used for scheduling tasks efficiently.

Chromosome Representation

Each chromosome represents one possible task schedule.

Example:

Tasks:

T1, T2, T3, T4

Chromosome:

[T2, T1, T4, T3]

This represents the execution order of tasks.

Fitness Function

The fitness function evaluates each schedule based on:

- Minimum completion time
- Resource utilization
- Cost
- Energy consumption

Higher fitness indicates a better schedule.

Selection

The best schedules with higher fitness values are selected as parents.

Crossover

Two parent schedules exchange task sequences to produce better offspring.

Example:

Parent 1:
T1 T2 T3 T4

Parent 2:
T3 T4 T1 T2

Offspring:
T1 T2 T1 T2

(After repair, duplicate tasks are removed to produce a valid schedule.)

Mutation

Mutation randomly changes the position of tasks.

Example:

Before Mutation:
T1 T2 T3 T4

After Mutation:
T1 T3 T2 T4

Mutation introduces diversity and avoids local optimum solutions.

Advantages

- Finds near-optimal schedules.
- Reduces execution time.
- Improves resource utilization.
- Handles complex optimization problems.

Conclusion

Selection, crossover, and mutation work together to evolve better task schedules, making Genetic Algorithms highly effective for optimization problems.

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15. Performance Evaluation of Classification Models (5 Marks)

Classification models are evaluated using different performance metrics to measure prediction quality.

Sample Confusion Matrix

Accuracy	**Predicted Positive**	**Predicted Negative**
Actual Positive	TP = 45	FN = 5
Actual Negative	FP = 10	TN = 40

Accuracy = (TP + TN) / Total

= (45 + 40) / 100

= **85%**

Accuracy measures the overall correctness of the model.

Precision

Precision = TP / (TP + FP)

= 45 / (45 + 10)

= **81.82%**

Precision measures how many predicted positive cases are actually positive.

Recall

Recall = TP / (TP + FN)

= 45 / (45 + 5)

= **90%**

Recall measures how many actual positive cases are correctly identified.

F1-Score

F1 = 2 × (Precision × Recall) / (Precision + Recall)

= **85.71%**

F1-score balances Precision and Recall.

ROC-AUC

The **Receiver Operating Characteristic (ROC)** curve compares the True Positive Rate with the False Positive Rate at different thresholds. The **Area Under the Curve (AUC)** indicates the model's ability to distinguish between classes. A value closer to **1** indicates excellent performance.

Real-World Applications

Spam Detection

- High Precision ensures legitimate emails are not incorrectly marked as spam.

Disease Diagnosis

- High Recall ensures that most patients with the disease are correctly identified, reducing missed diagnoses.

Conclusion

Evaluation metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC provide a comprehensive assessment of classification models. Choosing the appropriate metric depends on the application, ensuring reliable performance in domains such as healthcare, finance, and cybersecurity.